

The School Swimming team is looking to create a program that will determine the time it takes to swim 200m. Each swimmer can swim the 200m four times and the average of these four time is determined. To reduce the volume of input, these four times will be randomly generated. Based on a swimmer's average time, the swimmer is allocated to a team (A, B or Reserve). The program must also calculate the swimmer with the fastest average time.

2. **SwimTeam class** [6]

2.1 Create a Class called **SwimTeam**. (1)

2.2 In the program called **SwimTeam** add the following global variables that will be accessible throughout the program. Choose the appropriate types for each variable. (5)

name	Store text
teamListA	Store text
teamListB	Store text
teamListReserve	Store text
countA	Store whole number
countB	Store whole number
countReserve	Store whole number
aveTime	Store real number

3. **In the Main method** [6]

3.1 Create headings for the team lists by assigning "Team A" to **teamListA**, "Team B" to **teamListB** and "Team Reserve" to **teamListReserve**. Add a blank line after each heading. (3)

3.2 Add code to input a swimmer's name. (1)

3.3 Add code to call method called **oneSwim** to process the results on one swimmer. (2)

4. **In the oneSwim method** [39]

4.1 Add code to display the swimmer's name. Assuming the name is Thabiso, your output should look as follows: (2)

Swimmer Name: Thabiso

4.2 Write code to generate the minutes for the time which will be a random number between 2 and 4 (incl.). Assign this value to a variable called **swimMin**. (5)

4.3 Write code to generate the seconds for the time which will be a random number between 0 and 59 (incl.). Assign this value to a variable called **swimSec**. (2)

4.4 Using **swimMin** and **swimSec**, calculate the total time in seconds. Store this result in a variable called **swimTime**. For example, 2 min 55sec = 175 seconds. (3)

4.5 Write code to repeat steps 4.2 to 4.4 four times. (3)

4.6 Display the **swimMin**, **swimSec** and **swimTime** for each of the four generated times. Include the number for each time. An example of your output is shown below:

Swimmer Name: Thabiso

Swimmer Time 1: 3 min 0 seconds Total Time in Seconds is: 180 secs

Swimmer Time 2: 4 min 50 seconds Total Time in Seconds is: 290 secs

Swimmer Time 3: 3 min 53 seconds Total Time in Seconds is: 233 secs

Swimmer Time 4: 4 min 37 seconds Total Time in Seconds is: 277 secs

(4)

4.7 Add code to calculate the sum of the swimmers' four times. (4)

4.8 Add code to determine the average of the four times. (3)

4.9 Add code to assign the swimmer to a team, count the number of members in a team and add the swimmer's name to a team all depending on their average race time as follows:

4.9.1

Team A: average time is less than 160 seconds

Team B: average time is less than 210 seconds

Team Reserve: average time is 211 seconds and above (5)

4.9.2 Using the same programming structure, count the number of swimmers in each of the three teams. (3)

4.9.3 Using the same programming structure, add the swimmer's name to the appropriate team list, with a tab between each name. (3)

4.10 Display the **aveTime** for each swimmer in the following format. (2)

Average time in seconds: 245.0

4.11 Display the swimmer's allocated team. (1)

Team: Reserve

5 Return to the main method [14]

5.1 Create a while loop to input the names and process an unknown number of swimmers until "XXX" or "xxx" is entered. (4)

5.2 Add code to determine the lowest **aveTime**, and the name of the swimmer who achieved this time. (6)

5.3 Display the name of the swimmer with the fastest (lowest) time, using the following format:

The fastest swimmer was Thabiso with a time of 187.5

(2)

5.4 Call a method called **displayTeamList**. (2)

6 In the **displayTeamList** method [5]

6.1 Display a list of the teams, including the number of swimmers in each team. (5)

SAMPLE OUTPUT

Assuming you run the program for 10 swimmers called a, b, c, d, e, f, g, h, i, j; your output should appear similar to the following:

Swimmer Name: a	
Swimmer Time 1: 3 min 23 seconds	Total Time in Seconds is: 203 secs
Swimmer Time 2: 2 min 0 seconds	Total Time in Seconds is: 120 secs
Swimmer Time 3: 4 min 2 seconds	Total Time in Seconds is: 242 secs
Swimmer Time 4: 3 min 54 seconds	Total Time in Seconds is: 234 secs
Average time in seconds: 199.75	
Team: B	

Swimmer Name: b	
Swimmer Time 1: 3 min 20 seconds	Total Time in Seconds is: 200 secs
Swimmer Time 2: 4 min 48 seconds	Total Time in Seconds is: 288 secs
Swimmer Time 3: 4 min 7 seconds	Total Time in Seconds is: 247 secs
Swimmer Time 4: 3 min 0 seconds	Total Time in Seconds is: 180 secs
Average time in seconds: 228.75	
Team: Reserve	

Swimmer Name: c	
Swimmer Time 1: 4 min 29 seconds	Total Time in Seconds is: 269 secs
Swimmer Time 2: 4 min 20 seconds	Total Time in Seconds is: 260 secs
Swimmer Time 3: 4 min 8 seconds	Total Time in Seconds is: 248 secs
Swimmer Time 4: 3 min 23 seconds	Total Time in Seconds is: 203 secs
Average time in seconds: 245.0	
Team: Reserve	

Swimmer Name: d	
Swimmer Time 1: 2 min 15 seconds	Total Time in Seconds is: 135 secs
Swimmer Time 2: 2 min 16 seconds	Total Time in Seconds is: 136 secs
Swimmer Time 3: 3 min 48 seconds	Total Time in Seconds is: 228 secs
Swimmer Time 4: 3 min 56 seconds	Total Time in Seconds is: 236 secs
Average time in seconds: 183.75	
Team: B	

Swimmer Name: e	
Swimmer Time 1: 2 min 16 seconds	Total Time in Seconds is: 136 secs
Swimmer Time 2: 4 min 0 seconds	Total Time in Seconds is: 240 secs
Swimmer Time 3: 3 min 27 seconds	Total Time in Seconds is: 207 secs
Swimmer Time 4: 4 min 44 seconds	Total Time in Seconds is: 284 secs
Average time in seconds: 216.75	
Team: Reserve	

Swimmer Name: f	
Swimmer Time 1: 2 min 5 seconds	Total Time in Seconds is: 125 secs
Swimmer Time 2: 2 min 9 seconds	Total Time in Seconds is: 129 secs
Swimmer Time 3: 3 min 28 seconds	Total Time in Seconds is: 208 secs
Swimmer Time 4: 2 min 19 seconds	Total Time in Seconds is: 139 secs
Average time in seconds: 150.25	
Team: A	

Swimmer Name: g	
Swimmer Time 1: 2 min 9 seconds	Total Time in Seconds is: 129 secs
Swimmer Time 2: 4 min 37 seconds	Total Time in Seconds is: 277 secs

Swimmer Time 3: 2 min 27 seconds	Total Time in Seconds is: 147 secs
Swimmer Time 4: 4 min 17 seconds	Total Time in Seconds is: 257 secs
Average time in seconds: 202.5	
Team: B	

Swimmer Name: h	
Swimmer Time 1: 4 min 49 seconds	Total Time in Seconds is: 289 secs
Swimmer Time 2: 2 min 26 seconds	Total Time in Seconds is: 146 secs
Swimmer Time 3: 3 min 12 seconds	Total Time in Seconds is: 192 secs
Swimmer Time 4: 2 min 11 seconds	Total Time in Seconds is: 131 secs
Average time in seconds: 189.5	
Team: B	

Swimmer Name: i	
Swimmer Time 1: 3 min 9 seconds	Total Time in Seconds is: 189 secs
Swimmer Time 2: 2 min 12 seconds	Total Time in Seconds is: 132 secs
Swimmer Time 3: 2 min 17 seconds	Total Time in Seconds is: 137 secs
Swimmer Time 4: 2 min 8 seconds	Total Time in Seconds is: 128 secs
Average time in seconds: 146.5	
Team: A	

Swimmer Name: j	
Swimmer Time 1: 2 min 57 seconds	Total Time in Seconds is: 177 secs
Swimmer Time 2: 4 min 33 seconds	Total Time in Seconds is: 273 secs
Swimmer Time 3: 2 min 57 seconds	Total Time in Seconds is: 177 secs
Swimmer Time 4: 4 min 19 seconds	Total Time in Seconds is: 259 secs
Average time in seconds: 221.5	
Team: Reserve	

The fastest swimmer was i with a time of 146.5
The teams are:

Team A
f i
with 2 team member(s)

Team B
a d g h
with 4 team member(s)

Team Reserve
b c e j
with 4 team member(s)